

AY25 PROGRAM ASSESSMENT REPORT

v2026.1

Tracking Data

CIM code (e.g., ME-CM)	ME-ESEC
Program Name	Master of Engineering in Energy Systems Engineering, Energy Conservation and Buildings Track
Assessment Year	2024-2025
Program Assessment Coordinator	Brent Stephens
Year of Current Assessment Plan	2024-2025
No Assessment This Year per Plan	☐
LO(s) Assessed	2
Presented at program authority meeting date: <i>Please attach minutes</i>	n/a

Methodology

The Master of Engineering in Energy Systems, Energy Conservation and Buildings Track is a coursework-only graduate degree program oriented toward students who wish to develop more knowledge about the design, construction, and operation of buildings and their energy and environmental systems, as well as their integration with renewable energy systems and the electric grid. The Master of Energy Systems program has three tracks, each of which are assessed separately.

Learning objectives for this program are for students to:

1. Demonstrate advanced knowledge in and an ability to apply technical skills to fundamental problems in energy systems engineering, with a focus on energy conservation in buildings
2. Formulate research questions, conduct literature reviews, and utilize relevant tools to advance knowledge in energy systems engineering, with a focus on energy conservation in buildings

Learning Objective #1: Students will be able to demonstrate advanced knowledge in and an ability to apply technical skills to fundamental problems in energy systems engineering, with a focus on energy conservation in buildings.

- Artifact used to evaluate student learning:
 - Performance on assignments and exams in CAE 513 Building Science (using performance indicator ranging from 0 to 3)
 - Performance indicators are summarized topically into 4 key subject areas in architectural engineering: a) building physics, b) psychrometrics, c) ventilation and IAQ, and d) load calculations and energy balances
- Source for evaluation artifact (e.g., course(s), assignment(s), or other sources):
 - CAE 513 Building Science assignment and exam grades
 - Assignments and exams that cover:

- 1a: Building physics, heat transfer, and energy concepts (HW 1, HW 2, Exam 1)
 - 1b: Psychrometric principles and applications (HW 3, HW 4, Exam 2)
 - 1c: Ventilation and indoor air quality (HW 5)
 - 1d: Heating/cooling load calculations and energy balances (HW 6, Final Exam)
- Rationale for artifact and source:
 - The course is a key course in the program track requirements that all students are advised to take; it covers a breadth of fundamental subjects in energy systems engineering with a focus on energy conservation in buildings; each of the problem sets used for evaluation allow for systematic evaluation of performance indicators related to this LO
- Semester(s) in which artifacts are collected:
 - End of each academic year
- Sample size:
 - Each student in the program who took CAE 513 Building Science
- Data to report:
 - Results are reported as statistical distributions (i.e., histograms) and summary statistics for the sample of students in each year on each performance indicator
- Month/year of assessment:
 - End of each academic year
- Names/titles of evaluators:
 - Academic Program Coordinator, Program Director, and Department Chair

Learning Objective #2: Students will be able to formulate research questions, conduct literature reviews, and utilize relevant tools to advance knowledge in architectural engineering.

- Artifact used to evaluate student learning:
 - Performance on final written report for CAE 513 Building Science (using performance indicator ranging from 0 to 3)
- Source for evaluation artifact (e.g., course(s), assignment(s), or other sources):
 - CAE 513 Building Science final written report instructor evaluation
- Rationale for artifact and source:
 - The course is a key course in the program track requirements that all students are advised to take; it covers a breadth of fundamental subjects in energy systems engineering with a focus on energy conservation in buildings; written report allows for exploration of depth of knowledge and ability to use relevant tools within a single subject area related to energy conservation in buildings
- Evaluation rubric:
 - Clear justification and explanation of your ideas and motivation (20%)
 - Comprehensive literature review and quality/quantify of references (20%)
 - Follows specified format (10%)
 - Technical foundation for your project (20%)
 - Depth of investigation (30%)
- Semester(s) in which artifacts are collected:
 - End of each academic year
- Sample size:
 - Each student in the program who took CAE 513 Building Science
- Data to report:

- Results are reported as statistical distributions (i.e., histograms) and summary statistics for the sample of students in each year on each performance indicator
- Month/year of assessment:
 - End of each academic year
- Names/titles of evaluators:
 - Academic Program Coordinator, Program Director, and Department Chair

Results

Data table

The program had only 2 students in Fall 2024 and none took CAE 513 (30 graduate students in total took CAE 513 from a variety of majors). Therefore, this cohort lacks data for assessment.

Description of findings

Not applicable – no data for assessment in this cohort.

Prior CI plan impact

In the previous assessment report, there was only 1 student in the program who took CAE 513 (with no concerns). Prior CI plan impact is not applicable in this cohort without assessment data.

This cycle's CI plan

1. Given that there were two students enrolled in the program but none took CAE 513, deeper investigation revealed that 1 student had already taken CAE 513 in a previous year and had already been included in assessment, while another enrolled in courses but never showed and eventually withdrew.
2. Program track enrollment will continue to be monitored and evaluated.
3. Program advisors will continue to keep a close eye on students, especially for those coming from non-engineering backgrounds (per previous years' findings).
4. Continuing with previous years' plans and experiences with a parallel program that shares many courses with this track (ME-ARCE), advisors will attempt to improve preparation by students from non-engineering backgrounds by incorporating stricter prerequisite requirements at admission. We are paying closer attention to transcripts in the application review process, including scrutinizing what "engineering" means at different universities across the globe and helping those with non-engineering backgrounds transition into the program. In these cases, we now accept with deficiency course requirements of CAE 208 Thermodynamics and/or CAE 302 Fluid Mechanics, plus any required engineering mathematics courses that are missing.
 - a. None of the students in this cohort needed these prerequisite deficiency courses so this point is irrelevant for these students.